Shayantan Biswas
Bachelor of Technology
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GitHub Profile
LinkedIn Profile
LeetCode Profile

Education

•National Institute of Technology Durgapur *CGPA: 8.69*
Bachelor of Technology in Computer Science and Engineering 2023 – Present

Achievements

- Practices Data Structures and Algorithms daily through LeetCode to strengthen problem-solving and algorithmic thinking skills.
- Participated in **ISRO Hackathon**, **Walmart Hackathon**, and **Smart India Hackathon (SIH) 2024 , 2025**, showcasing teamwork, innovation, and problem-solving skills.
- Selected for **Amazon ML Summer School 2026** through Unstop, gaining the opportunity to learn advanced Machine Learning concepts from Amazon industry experts.
- Successfully advanced to the **3rd Round of Adobe India Hackathon 2026**, demonstrating strong problem-solving, programming, and technical skills in a competitive national-level hackathon.

Personal Projects

• Full Stack Food Application with AI Integration

A full-stack web application for browsing, classifying, and ordering food using advanced AI features.

- Integrated a voice-enabled AI agent for website navigation, personalized food recommendations, cart management, and order placement using natural-language commands.
- Developed an image-based food classification system using a custom-trained deep-learning model with a Flask backend.
- Implemented secure authentication, menu browsing, dynamic cart functionality, order management, and chatbot-based interaction.
- **Tech Stack:** React.js, Node.js, Express.js, Flask, Tailwind CSS, MongoDB, Dialogflow, Web Speech API.
- **Repository:** GitHub Link

• GATEMIND-AI

An AI-powered and personalized platform for GATE preparation and performance analysis.

- Developed JWT authentication, OTP verification, mock tests, and subject-wise performance analytics.
- Implemented a personalized RAG chatbot with hybrid semantic–lexical retrieval, document embeddings, conversation history, and citation-based responses.
- Designed a modular Flask backend using SOLID principles and multiple low-level design patterns.
- Deployed the React frontend on Vercel and the Dockerized backend on AWS EC2 using GitHub Actions CI/CD.
- **Tech Stack:** React.js, Flask, MongoDB, LangChain, RAG, Docker, AWS, GitHub Actions.
- **Repository:** GitHub Link

• Explainable AI for Plant Seedling Classification

Deep-learning-based plant seedling classification using Explainable AI techniques.

- Conducted a research project on plant seedling image classification using multiple deep-learning architectures.
- Applied LIME and SHAP to interpret model predictions and analyze influential image regions.
- Evaluated classification performance using standard evaluation metrics and visualization techniques.
- **Tech Stack:** PyTorch, LIME, SHAP, Scikit-learn, Matplotlib, Pandas.
- **Repository:** GitHub Link

Experience

•CMATERLab, Jadavpur University *Dec 2024 – Jan 2025*

Winter Research Intern – Deep Learning for PV Cell Generation Forecasting Remote

– Worked under Prof. Debotosh Bhattacharjee on photovoltaic power forecasting using deep learning techniques.
– Applied feature selection methods and ensemble models on time-series datasets, achieving **98% forecasting accuracy**.

•Indian Institute of Technology, Guwahati *May 2025 – Jul 2025*

Summer Research Intern – Tabular Deep Learning, Self-Supervised Learning Remote

– Worked under Prof. Arijit Sur on evaluating self-supervised learning approaches for structured/tabular datasets.
– Implemented and benchmarked multiple deep learning models, achieving up to **70% accuracy**.

•Indian Institute of Technology, Hyderabad *May 2026 – Jul 2026*

Summer Research Intern (SRFP) – Clustering and Multithreading On-Site

– Worked under **Prof. Sathya Peri** on incremental clustering techniques, implementing and optimizing the **BIRCH** algorithm in **C++** for efficient, scalable, and incremental data clustering.

Technical Skills

Programming Languages: C/C++, Python, JavaScript, HTML, CSS
Frameworks: React.js, Express.js, Flask
Libraries: NumPy, Pandas, Matplotlib, PyTorch, TensorFlow, C++ STL
Web Development Tools: Node.js, VS Code, Git, GitHub
Cloud & DevOps: AWS EC2, AWS IAM, AWS S3, Docker
Databases: MongoDB, MySQL
AI & Machine Learning: Machine Learning, Deep Learning, Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG)
Core CS Concepts: Data Structures & Algorithms, Operating Systems, Object-Oriented Programming, Database Management Systems, Software Engineering

Leadership and Responsibility

Head of Data Science & Treasurer *Feb 2025 – Present*
IEEE Computer Society, National Institute of Technology Durgapur

– Led the planning and execution of flagship technical events including **Hack Hustle** (24-hour hackathon), **ENVISION** (autonomous bot challenge), and **PCB Decoded**, driving student engagement and technical learning.
– Conceptualized and organized **Epoch**, a **5-day AI/ML workshop and hackathon** for first-year students, sponsored by **GeeksforGeeks (GFG)**, introducing participants to machine learning fundamentals and practical development workflows.